

# Mohamed Ilyes Tounsi

FULL-STACK SOFTWARE ENGINEER · AI/ML INTEGRATOR · FOUNDER

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## Professional Summary

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Full-stack software engineer and engineering leader with 20+ years of experience shipping production web platforms end-to-end — architecture, backend APIs, cloud infrastructure, frontend, data pipelines, and CI/CD. Over the last three years I've layered **AI/ML integration** on top of that foundation: LLM applications, RAG pipelines, MCP-based agent orchestration, voice, semantic search, and human-in-the-loop workflows. Current focus splits across a fractional senior engineering role at NXT Robotics (~60%), an advisory VP-of-Engineering role at AIMIA (~20%), part-time engineering at Hydrostasis (~20%), and solo-founder work on [digitalqrcard.com](https://digitalqrcard.com).

I'm a **software engineer first**. AI is a tool in the stack, not the whole stack.

## Core Skills

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**Languages & Frameworks** — TypeScript, JavaScript, Python, PHP, Go (reading), Node.js, React 18, Next.js 15, Vite, FastAPI, Express, Hono, Fastify, Laravel, GraphQL, REST.

**Frontend** — React, Next.js, Vite, Tailwind CSS, Radix UI, Chakra UI, TanStack Router/Query, styled-components, CustomTkinter (desktop), responsive design systems.

**Backend & Cloud** — Node.js, Python (FastAPI, Django), Docker, Kubernetes, AWS (ECS, EC2, S3, CloudFront, Route 53, Lambda, CloudFormation, EventBridge, Secrets Manager), Vercel, Railway, Neon, Supabase, GitHub Actions.

**Data & Storage** — PostgreSQL, MongoDB, ChromaDB, Firestore, SQLite, Drizzle ORM, Prisma, ETL pipelines, data modeling, vector embeddings.

**AI/ML Integration** — Anthropic Claude SDK, OpenAI, Google Vertex AI / Gemini, Groq, ElevenLabs voice, HuggingFace embeddings, MCP (Model Context Protocol) orchestration, RAG pipelines, semantic search, human-in-the-loop workflows, weak-supervision labeling.

**Testing & Quality** — TDD, Jest, PHPUnit, Playwright, Autocannon, unit / integration / E2E.

**Delivery & Leadership** — Agile/Scrum, Jira, ClickUp, Asana, distributed team leadership, mentoring, code review at scale.

## Experience

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### Senior Software Engineer — NXT Robotics (Remote) · Fractional (~60%) · 2025 – Present

- Improved frontend, backend, and APIs powering robot control, monitoring, diagnostics, and command dashboards.
- Integrated AI/ML into robotic decision-making pipelines.
- Multi-module architecture: React dashboard, Python deploy tooling, Node workers, financial dashboard, internal "nxt-brains" services.
- **Stack:** React + Ant Design + Redux + Chart.js, Python, Node, ElevenLabs, Laravel, PHP, AWS, Docker, REST APIs.

## **VP of Engineering (Advisory, Part-time ~20%) — AIMIA (Remote) · San Diego, CA · 2025 – Present**

- Advise on engineering strategy for a distributed software, platform, QA, and data team.
- Architect and oversee an AI-powered career advisory platform: FastAPI + Node backend, React + Next.js frontend, MongoDB, AWS.
- Establish CI/CD pipelines, deploy on AWS (ECS, EC2, Route 53), and maintain production reliability.
- Mentor engineers; instill disciplined execution across Agile teams.
- **Stack:** Node.js, Python, FastAPI, Express, Next.js, Tailwind, MongoDB, GitHub Actions, AWS, Docker.

## **Software Engineering Manager — Zembra (Remote) · San Diego, CA · Sep 2024 – Nov 2024**

- Led a fullstack team of 6 delivering core platform features Agile-style.
- Architected web apps and REST APIs using Node.js, React, PHP, SQL.
- Scrum Master; facilitated sprints and cross-team collaboration.

## **GIS Enterprise Architect — geoConvergence (Remote) · Bloomington, IN · 2023 – 2024**

- Architected high-availability GIS solutions and microservices on AWS (Lambda, CloudFormation, EventBridge).
- Ensured reliability, scalability, and secure deployment of data pipelines and GIS applications.
- **Stack:** AWS, PostgreSQL, Esri ArcGIS, Docker, CloudFormation, EventBridge.

## **Colonel & Intelligence Fusion Center Manager — Tunisian Ministry of National Defense · 2001 – 2022**

- Led software engineering efforts as a senior manager: architecture, delivery, team performance for mission-critical systems.
- Built scalable PHP-based platforms for eCommerce and marketing.
- Directed intelligence data consolidation into a real-time analytics dashboard with interactive visualizations ("Intelligence Data Lake").
- **Stack:** PHP, MySQL, Node.js, Leaflet.js, REST APIs, Docker, GitHub.

## **Featured Project — Digital QR Card ([digitalqrcard.com](https://digitalqrcard.com))**

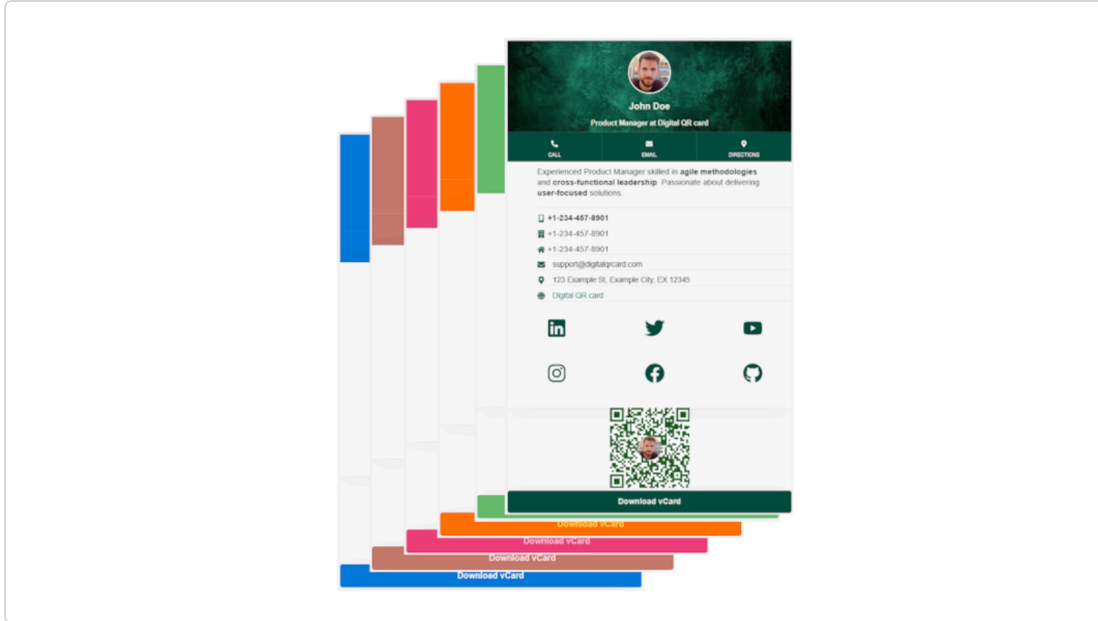
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**Solo founder & operator.** B2C business selling customizable QR-based digital business cards with **built-in lead capture** — the differentiator vs Popl / HiHello / Linq / Blinq / Mobilo, all of which treat the card as the product. We treat the *connection* as the product.

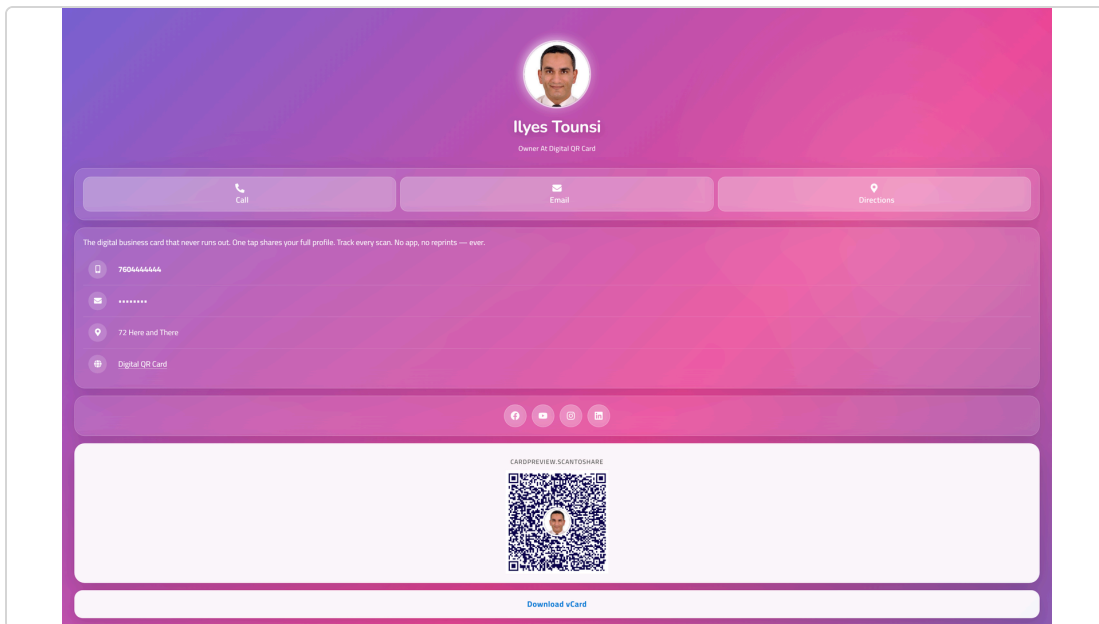
**Master hook:** *"Every scan becomes a lead."* **Primary audience (last 90 days):** Real estate agents — open houses, listing presentations, walk-through follow-up.

### **The product**

Every card is fully customizable — theme, color, layout, socials, QR — and lives at a persistent URL so the info never goes stale.



The scan target: a live, mobile-first profile card with call / email / directions actions, socials, vCard download, and (the moat) an active lead-capture form.

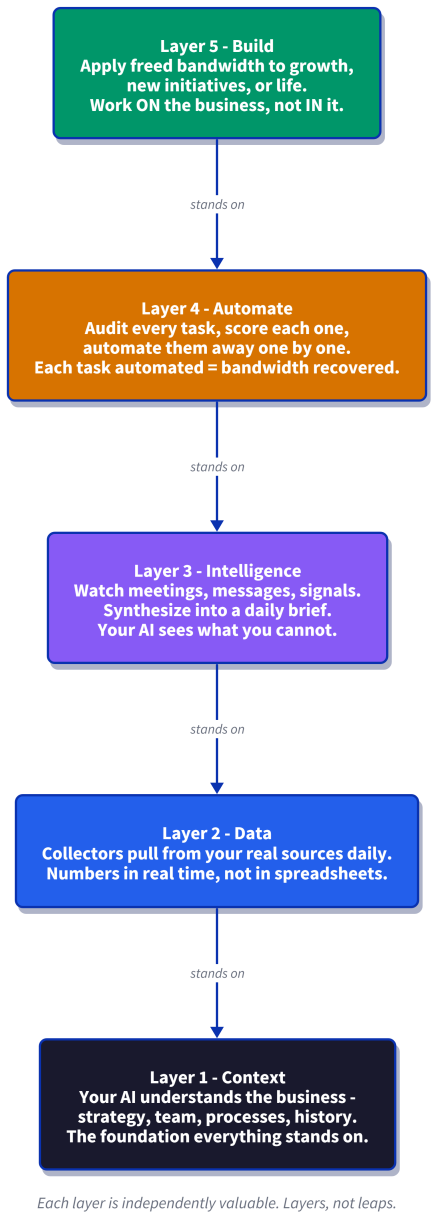


## System architecture

The platform is more than the product page — it's a modular **AI Operating System (AIOS)** wrapped around the business. Five layers, installed one at a time:

# AIOS - Five Layers

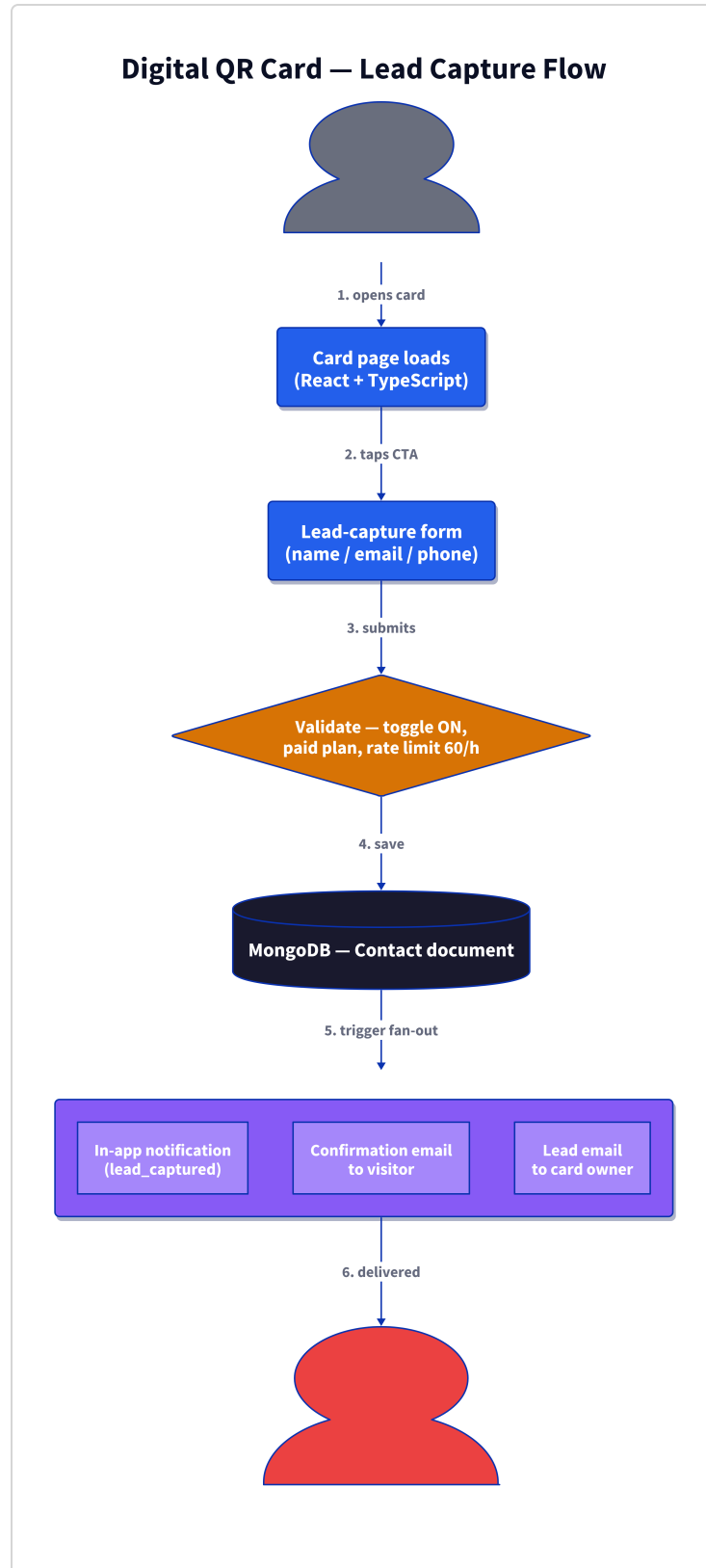
An AI operating system wrapped around your business - built one layer at a time



| Layer        | Responsibility                                                                                                                        |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Context      | Machine-readable business truth (strategy, offers, brand, personal info). Every AI session boots with <code>/prime</code> to load it. |
| Data         | Daily collectors pull real numbers from analytics, Stripe, Shopify, ad platforms.                                                     |
| Intelligence | Meetings, messages, and signals synthesized into a daily brief.                                                                       |
| Automate     | Every recurring task audited, scored, and automated one at a time.                                                                    |
| Build        | Freed bandwidth applied to growth, product, and net-new initiatives.                                                                  |

## Lead-capture flow — the moat

Every scan of a Digital QR Card lands on a personalized profile with an *active* lead-capture form (vs the passive vCard download every competitor ships). The lead flows straight into the owner's inbox and CRM.



## What I built (technical)

- **Product surface:** React + TypeScript customizable card builder; QR renderer; profile page with lead-capture form; Stripe checkout; MongoDB-backed profiles.
- **AIOS runtime:** Claude-Code-driven modular workspace — slash commands ( `/prime` , `/install` , `/create-plan` , `/implement` , `/share` , `/deck` , `/youtube-screencast` ) that operate on context files, plans, and outputs stored in the repo.
- **Diagram engine:** D2 source files rendered to PNG for docs, decks, and this résumé — same PNGs embedded above.
- **Content pipeline:** YouTube-first content system, 9 pieces/week across LinkedIn / IG / TikTok / Facebook, with a `writing-style` skill enforcing anti-AI-slop copy.
- **Deck builder:** Marp-based Markdown-to-PPTX pipeline branded to digitalqrcard.com.
- **Guardrails:** `human-in-the-loop` by default , `local data` by default , `borrow before you build` — encoded as workspace principles, not aspirational slogans.

## What I built (business)

Solo across product, marketing, sales, ops. Currently in the customer-acquisition sprint: reach \$8,000/month within 90 days. Everything upstream (product, pricing, packaging) already stabilized; the binding constraint is monthly visitors, which the content pipeline is designed to unblock.

Founding member of the **AAA Accelerator** cohort ([aaaaccelerator.com](https://aaaaccelerator.com)) — the AI business launch & AIOS program that seeded this architecture.

## Featured Client Engineering

### NXT Robotics — Unified robotics operations dashboard

Multi-module system consolidating robot control, monitoring, diagnostics, financial ops, and voice-driven "cockpit" interaction into a single operations surface for a robotics company. Ongoing senior engineering role; cockpit demo in flight.

- **Sub-modules I contribute to:** `dashboard-react-js` (operator UI), `dashboard-api` (backend), `dashboard-deploy` (Python deployment tooling), `P-NXT-financial-dashboard` (revenue/ops view), `nxt-brains` (internal service).
- **What's interesting:** heterogeneous stack unified behind one operator experience; voice channel via ElevenLabs; Chart.js-backed real-time telemetry; Redux state across long-running dashboards.
- **Stack:** React 18, Ant Design, Redux, Chart.js, Node, Python, Laravel, PHP, ElevenLabs, AWS, Docker, REST APIs.

### Hydrostasis — Retool admin portal with Test → Prod promotion pipeline

Admin portal for Hydrostasis (intern summer 2026 engagement). Phase 1 promoted to production; phase 2 (webhook integrations) in flight.

- **What's interesting:** wrote `promote.js` — a scripted Test → Prod promotion workflow for Retool app definitions, so deployments are diff-reviewable and reversible instead of manual copy-paste (which is the Retool default and a common source of prod breakage).
- Modern Vite / TanStack Router / TanStack Query / Radix UI stack layered over Retool's admin primitives.
- **Stack:** Node, Vite, TanStack Router/Query, Radix UI, Retool.

### AIOS — AI-voice QR card + realtor pilot

The engineering repo behind [digitalqrcard.com](https://digitalqrcard.com) (see hero project above). Realtor B2B validation sprint underway with Jake Rose pilot.

- **Stack:** Node, Express, MongoDB, ElevenLabs, Stripe.

### SEAM — Synthetic Intelligence Training Data Generator

"Plan first. Generate second. Provenance receipt attached." A local Node application that produces defensible synthetic training packets for intelligence-analyst instructors — the classified case files real analysts learn from are, by definition, unteachable, so this generates realistic non-classified equivalents from four card taps and about ninety seconds of compute.

- **Five-stage pipeline ( world → artifacts → verifier → provenance → packet ):**
  1. Claude Sonnet builds the underlying truth of the case as a structured JSON world model (entities, timeline, evidence chain, answer key). Zod-validated.
  2. Claude Sonnet *derives* each artifact (cables, banking records, social posts, intercepts) from that world — nothing invented at this layer.
  3. Claude Haiku reads every artifact back against a versioned **seam taxonomy** and flags temporal / geographic / identity / financial / linguistic contradictions.
  4. A machine-readable **provenance receipt** names the exact models used, taxonomy version, guardrails applied, and human sign-off status.
  5. A single PDF assembles cover + artifacts + receipt — the printed packet the instructor hands to the class.
- **Why the shape matters:** naive multi-document LLM generation drifts — ask one prompt for 50 artifacts and by artifact 8 the model has forgotten what it said on artifact 3. Committing truth to a world model first, then *deriving* rather than re-generating, kills that drift. The Haiku verifier catches whatever survives.
- **Two front doors on the same pipeline:** a Fastify web UI (card-stack interface the instructor drives) and a `seam` CLI (interface the developer drives) — both call the same orchestrator.
- **Target users:** federal intelligence schoolhouses (primary), academic programs (Mercyhurst, AMU, MiraCosta CSIT), Five Eyes partner training, corporate threat-intel onboarding.
- **Stack:** TypeScript (strict), Node, Fastify, Anthropic Claude Agent SDK (Sonnet + Haiku), Zod, YAML case configs, PDF assembly.

### hitl-social — Human-in-the-loop social monitoring & lead approval

Python service that monitors social channels, uses an LLM to score / classify potential leads, and routes candidates through a Slack approval flow before any outbound action. LinkedIn variant in a sibling repo.

- **What's interesting:** interactive Slack Block Kit UI for approve / reject / edit; explicit HITL boundary — the LLM never posts autonomously.
- **Stack:** Python, FastAPI, Groq LLM, Slack SDK.

### PurifiedFeed — Content-moderation monorepo

Turborepo platform combining a Next.js dashboard, an Express ingestion service, Firestore storage, and Google Gemini for content classification.

- **Stack:** Turborepo, Next.js 15, Express, Firestore, Google Gemini, TypeScript.

## Archestra fork (via P-algora-bounties) — MCP orchestrator contribution

Contributor to an MCP-native governance / orchestration platform (Go backend, Docker/Kubernetes). Companion Python `bounty-watcher` script tracks Algora open bounties.

- **What's interesting:** production-grade MCP orchestrator surface — real-world exposure to MCP server auth, tool discovery, and multi-tenant governance patterns.

## RevenueSignalAI — Multi-tenant lead-gen automation platform

A two-product suite plus its own marketing surface, aimed at converting external signals (city permits, YouTube niches, social) into qualified sales leads for SMB clients.

- `automation-app` — Multi-tenant YouTube content-automation platform. Express / Prisma backend + Next.js frontend, with JWT-authenticated admin panel for tenant/user provisioning, Google/YouTube OAuth per tenant, encrypted OAuth tokens at rest (32-byte base64 key), and topic-to-short generation via **Google Gemini**. Each tenant gets isolated channels, credentials, and post history.
- `BarnesCo_Permit_Monitor` — Production Python scraper deployed for a real client (Barnes & Co). Polls City of San Diego + San Diego County public permit databases every 15 minutes for senior-housing permits (assisted living, memory care, nursing facilities), emails matched leads to the sales team via Gmail App Password, and appends to a CSV with `contacted` / `notes` columns so the sales team tracks follow-up directly in the sheet.
- `RevenueSignalAI.com` — Static Next.js marketing site.
- **What's interesting:** encrypted-at-rest OAuth token storage; per-tenant credential isolation; the permit-monitor is a "boring but works" counterexample to over-engineering — one Python file, cron loop, actual leads.
- **Stack:** Next.js 15, Express, Prisma, PostgreSQL / Supabase, TypeScript, Google Gemini, YouTube Data API, Python, cron. Web on Vercel, worker on Railway.

## Snorkel task-authoring factory

In-flight task-authoring pipeline for domain-specific ML labeling competitions. Focus on repeatable task specs, weak-supervision heuristics, and rapid iteration on label functions.

## Featured Full-Stack Projects (SWE-heavy)

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### DigitalQRCard-FullStack — Predecessor architecture

The pre-AIOS version of what became [digitalqrcode.com](https://digitalqrcode.com). Full React/TypeScript frontend + Node/TypeScript backend + Docker + MongoDB. Superseded but demonstrates the domain trajectory and end-to-end delivery pattern.

### Dropshipping platform

Next.js 15 + Prisma + next-auth 5 e-commerce with Cheerio-based scraping and Groq LLM enrichment; admin user provisioning scripts.

### HomeServiceHub — Booking / management app

React + Express + Radix UI + Drizzle ORM + Neon serverless Postgres. Vite + TypeScript build; esbuild-bundled server.

### Timecards — Local time tracker (shipped)

Python + tkinter + SQLite desktop app; v1.0 released, v1.1 planned (idle detection, code-signing, Microsoft Store distribution). Tray badge, soft-delete, overlap detection, light/dark theme.



## GreenGearCA — Eco hiking blog + shop

Node + React + Tailwind, Markdown-driven blog engine.

## tn76.com portfolio & satellite sites

Personal portfolio (TypeScript + React 18 + Vite + Chakra UI + Recharts) plus a family of satellite properties on subdomains — radio/TV streaming ( [P-tn76-radiotv](#) ) and AI-post-gen-to-Blogger syndication ( [P-trends.tn76.com](#) ).

## PurifiedFeed / Reddit Devvit / Video-editor-python

See personal AI/ML section for full-stack pieces that also demonstrate frontend + tooling breadth.

## Meta Project — Personal Portfolio Index & Contractor Income System

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Because the portfolio spans ~40 concurrent projects and multiple income streams, I built a lightweight **AIOS-lite for myself**: a Markdown-based operating layer that keeps the whole picture coherent across Claude Code sessions and human-only editing.

- **PROJECTS-INDEX.md** — human-editable source of truth. One table per group (Client / Org / Personal-active / Personal-warm / Personal-dormant / Other) with purpose, stack, status, last-touch, and cross-references to income streams.
- **CONTRACTOR-INCOME/** — weekly income rollup (INCOME-TRACKER.md) across Snorkel, Algora, Mercor, DataAnnotation, Scale, ClifyX/Meta + a recurring Monday MERCOR-CHECKLIST.md with interview-prep notes.
- **Memory mirror** at `~/claude/projects/c--dev/memory/projects_overview.md` — auto-loaded into every Claude Code session under `C:\dev`, so any project session inherits awareness of the other 39.
- **Update discipline**: trigger phrase ( `"update the projects index"` ) refreshes both index + memory in the same turn; Friday ritual captures anything I forgot to log during the week.

Small project, but it's the reason all the above stays organized enough to ship.

## Contract & ML Data Work

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Active contractor across ML data, weak-supervision, and bounty platforms — informs the featured LLM/RAG work above.

- **Snorkel AI** — task authoring for domain-specific labeling factories.
- **Algora** — open-source bounties (see *Archestra* above).
- **Mercor** — active on the Instant Offers pipeline.
- **DataAnnotation, Scale AI, ClifyX / Meta** — LLM data annotation, evaluation, and red-teaming assignments.

## Current Client Engagements (Freelance)

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- **Hydrostasis** — Part-time (~20%) Retool admin portal engineering (summer 2026). Node / Vite / TanStack / Radix. Phase 1 promoted to prod; phase 2 (webhooks) in flight. Includes a scripted Test → Prod promotion workflow ( `promote.js` ).
- **Individual SWE contracts** — Ongoing engineering-for-hire work for private clients. Recent example: loading a 49-software PLR bundle into an existing self-hosted OpenCart store with automated digital delivery, Stripe + PayPal integration, and a product-page redesign to match a reference storefront.
- **GBDT** — Deliverables and research assets around gradient-boosted-decision-tree workflows.
- **geoSecureTech** — React 18 + Vite + Bootstrap 5 dev site with i18n.

## Education

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- **M.S. in Systems Technology** — U.S. Naval Postgraduate School, 2012
- **M.Eng. in Remote Sensing & GIS** — Université Pierre & Marie Curie, 2002
- **B.Eng. in Telecommunications** — Aviation School of Borj El Amri, 2001

## Training

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- Critical Thinking — NATO School, 2018
- Strategic Intelligence Leadership Course — U.S. Defense Intelligence Agency, 2018
- Geographic Information Systems — U.S. Federal Bureau of Investigation, 2017
- Intelligence, Surveillance & Reconnaissance — Torchlight (UK), 2017
- Big Data, Data Science, ML, IoT — EuraNova Experts, 2016
- Geo Database Management & Maintenance — U.S. National Geospatial Intelligence Agency, 2016
- **AAA Accelerator (2026)** — AI business launch & AIOS program (basis for the Digital QR Card architecture above).

## Languages

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- **English** — Fluent
- **French** — Fluent
- **Arabic** — Native